25901556 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCSAC200 - HEART RATE TRAINING

(2024 ADMISSION ONWARDS)

Answer scheme

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. b) Gradually increase the training demand
2. a) Increased ventricular filling
3. d) Blood is redirected to the skin for cooling
4. b) Assess long-term adaptation to training
5. c) Pump blood
6. b) Mediastinum
7. c) Peripheral resistance
8. a) Higher metabolic rate
9. d) Prepare muscles and cardiovascular system for activity
10. d) Maximal graded exercise test. **[1x10 = 10]**

Part B

Short Answer

Answer any **10** questions

Each question carries **2** marks

11. Define Afterload
    - Afterload is the pressure or resistance the ventricles must overcome to eject blood during systole.
    - It is mainly determined by arterial blood pressure and vascular resistance.
12. How does an increase in vascular resistance affect blood pressure?
    - Increased vascular resistance makes it harder for blood to flow through vessels.
    - This increases arterial blood pressure because the heart must pump against greater resistance.
13. State the relationship between heart rate and oxygen consumption during exercise.
    - Heart rate and oxygen consumption (VO_2) increase proportionally during exercise.
    - As exercise intensity rises, muscles require more oxygen, causing heart rate to increase.
14. Explain walking training zones and their significance in exercise prescription.
    - Walking training zones are specific heart-rate ranges used during walking exercise.
    - They help control exercise intensity and improve fitness safely and effectively.
    - They assist in achieving goals such as weight loss, endurance, and cardiovascular health.
15. Explain the importance of ventricular diastole in the cardiac cycle.
    - Ventricular diastole is the relaxation phase of the ventricles.
    - It allows ventricular filling with blood from the atria.

- Adequate filling ensures an optimal stroke volume during the next contraction.

16. How does increased afterload influence stroke volume?
    - Increased afterload makes it more difficult for the heart to eject blood.
    - As a result, stroke volume decreases if cardiac contractility does not increase.
17. Calculate the maximum heart rate of a 20-year-old.
    - Formula: Maximum Heart Rate = 220 – Age = 220 – 20

    Maximum Heart Rate = 200 beats/minute (bpm)
18. Differentiate between continuous training and interval training.

    1. Continuous Training
    - Exercise is performed continuously without rest.
    - Usually done at a steady, moderate intensity.
    - Primarily improves aerobic endurance.
    - Causes less fatigue compared to interval training.
    - Example: Continuous jogging or cycling.

    2. Interval Training
    - Exercise is performed with alternating work and rest periods.
    - Usually done at moderate to high intensity.
    - Improves both aerobic and anaerobic fitness.
    - Causes greater fatigue due to higher intensity.
    - Example: Sprinting followed by walking recovery periods.

19. How does training in the anaerobic zone differ from training in the aerobic zone?
    - Anaerobic zone training involves high-intensity exercise with limited oxygen availability.
    - Aerobic zone training involves moderate-intensity exercise where oxygen supply meets demand.
    - Anaerobic training improves power and speed, while aerobic training improves endurance.
20. Explain the role of venous valves in venous return.
    - Venous valves prevent backflow of blood.
    - They ensure one-way movement of blood toward the heart.
    - They enhance venous return, especially during muscle contractions.

21. Why is plasma volume important for cardiovascular efficiency during exercise?

- Plasma volume maintains blood volume and blood pressure.
- It improves venous return and stroke volume.
- It helps regulate body temperature and transport oxygen and nutrients.

22. Define heart rate training zone.

- A heart rate training zone is a specific range of heart rates used to achieve a particular fitness goal.
- It is usually expressed as a percentage of maximum heart rate. **[2x10 = 20]**

Part C

Short Essay

Answer any **4** questions

Each question carries **5** marks

23. Define energy expenditure and explain its relationship with heart rate during exercise.

- Energy Expenditure: The amount of energy used by the body to perform physical activity and maintain bodily functions.
- Relationship with Heart Rate:
- Heart rate increases as exercise intensity increases.
- Higher intensity requires greater energy expenditure.
- Increased oxygen demand causes the heart to pump faster.
- Therefore, heart rate is often used to estimate energy expenditure.
- There is a near-linear relationship between heart rate and oxygen consumption during moderate exercise.

24. Explain how electrical impulses are generated and conducted through the heart.

- Electrical impulses originate in the Sinoatrial (SA) Node, the heart's natural pacemaker.
- The impulse spreads through both atria causing atrial contraction.
- It reaches the Atrioventricular (AV) Node, where a brief delay occurs.
- The impulse then travels through the Bundle of His.
- It continues through the right and left bundle branches.
- Finally, Purkinje fibers distribute the impulse throughout the ventricles.
- This coordinated conduction produces synchronized ventricular contraction.

25. Explain the methods used to measure VO_2 max and its importance.

1. Methods of Measuring VO_2 Max

- Direct Method
- Measured in a laboratory using a treadmill or cycle ergometer.
- Oxygen uptake is analyzed using a metabolic gas analyzer.

2. Indirect Methods

- Cooper 12-Minute Run Test.
- Queen's College Step Test.
- Rockport Walking Test.
- Astrand Cycle Ergometer Test.

3. Importance of VO_2 Max

- Indicates aerobic fitness level.
- Reflects the body's ability to transport and utilize oxygen.
- Useful for athlete evaluation and training prescription.

- Predicts endurance performance.
- Monitors training progress.

26. Describe the factors affecting ventricular filling time.
    - Heart Rate: Increased heart rate reduces filling time.
    - Venous Return: Greater venous return increases ventricular filling.
    - Atrial Contraction: Enhances ventricular filling.
    - Ventricular Compliance: More compliant ventricles fill more easily.
    - Blood Volume: Increased blood volume improves filling.
    - Duration of Diastole: Longer diastole allows greater filling.
27. Define cardiac output. Explain the formula and normal resting values.
    - Cardiac output is the volume of blood pumped by the heart per minute.
    - Formula CO = HR × SV
    - CO = Cardiac Output
    - HR = Heart Rate
    - SV = Stroke Volume
    - Normal Resting Values
    - Average resting heart rate = 70–75 bpm.
    - Average stroke volume = 70 mL/beat.
    - Normal resting cardiac output ≈ 5 L/min.
    - May increase to 20–40 L/min during intense exercise.
28. Analyze how endurance training improves aerobic performance.
    - Increases VO_2 max.
    - Enhances cardiac output through increased stroke volume.
    - Improves capillary density in muscles.
    - Increases mitochondrial number and function.
    - Improves oxygen extraction by working muscles.
    - Delays fatigue by enhancing aerobic energy production.
    - Increases endurance and exercise performance.
    - Improves efficiency of the cardiovascular and respiratory systems.

[5x4 = 20]